Leo Marcinkus

leomarcinkus@gmail.com | leo.marcinkus@colorado.edu | (630) 864-8802
linkedin.com/in/leo-marcinkus-a18062340 | github.com/Leo-Marcinkus/Engineering-Portfolio

Electrical engineering undergraduate at CU Boulder (accelerated B.S./M.S.) with hands-on experience in signal integrity, PCB design, embedded systems, and analog and RF circuit design, seeking to contribute to hardware and systems engineering work.

EXPERIENCE

Siemens EDA, Software Test Intern (HyperLynx QA) *Boulder, CO | May 2026 - Present*

- Execute and validate HyperLynx SI, PI, AMS, DRC, and Advanced Solver workflows, including DDR4 and DDR5 timing margin, eye diagram, and crosstalk analysis with IBIS models.
- Run power integrity simulations covering DC drop, target impedance, decoupling, and VRM behavior, and extract RLGC, current density, and EMI results with advanced solvers.
- Document reproducible software defects and workflow inconsistencies, contributing engineering reports and usability recommendations across releases.

Independent Study, RF Receiver Design (Prof. Bogatin) *CU Boulder | January - May 2026*

- Designed a digitally tuned AM receiver front end using a 7-bit switched capacitor bank (2 pF to 302 pF, 128 combinations) on an LC tank, with an Arduino and analog switch ICs replacing a mechanical variable capacitor.
- Validated a working prototype with a Digilent AD3, measuring resonance from 297 kHz to 561 kHz, characterizing 154 pF of stray capacitance from a 0 pF baseline sweep, and recovering AM audio to roughly 10 kHz.

PROJECTS

Golden Arduino Board

- Designed and assembled a custom multi-layer ATmega328 board in Altium, interpreting multiple datasheets and benchmarking it against a stock reference board with oscilloscope measurements.
- Measured up to **1.7x lower ground bounce**, **1.9x better on-die supply isolation**, **2.2x better isolation** from external aggressors, and **4.1x lower near-field emissions** than stock, from local decoupling placement, tighter return paths and current loops, and via stitching.

1T-1C DRAM Cell with Cross-Coupled CMOS Sense Amplifier

- Designed, simulated, and laid out a 13-transistor two-cell DRAM array in Siemens Tanner S-Edit, T-Spice, and L-Edit on a Generic 250 nm process at 2.5 V, including precharge, equalization, and SAE/SAEB timing.
- Measured a 57 mV pre-latch sense margin against a 50 mV target, confirmed amplification to a 1.5 V resolved differential, and verified destructive read and partial storage node refresh on both cells.

Retro Glass Fuse Lamp

- Built a retro-aesthetic LED lamp with a rechargeable battery circuit, ESP-NOW wireless control, and battery safety practices.

LABORATORY AND COURSEWORK EXPERIENCE

Embedded Software and RTOS (2025 - 2026): ARM Cortex-M firmware in C, FreeRTOS task design with semaphores and mutexes, tracing and profiling

PCB Design and Manufacturing (2025): multi-layer Altium PCBs, signal integrity, crosstalk, and power-rail noise measurement

Advanced Electronic and Semiconductor Lab (2025): BJT, diode, photodiode, and MOSFET characterization, I-V measurement, parameter extraction

TECHNICAL SKILLS

Hardware and PCB Design: Altium Designer, Siemens Tanner S-Edit and L-Edit, Cadence Virtuoso, PDN and power management

Signal and Power Integrity: HyperLynx (SI, PI, AMS, Advanced Solvers), IBIS and IBIS-AMI, eye diagrams, crosstalk, LTSpice, ADS

Embedded Systems: STM32 and ARM Cortex-M, FreeRTOS/CMSIS-RTOS2, Arduino, PWM, SPI/I2C/UART, RISC-V

Programming and Tools: C/C++, Python/MicroPython, SystemVerilog, Assembly, Git, Debian and Arch based Linux

Test and Measurement: Oscilloscopes, logic analyzers, DMM, waveform generators, network analyzer sweeps, I-V characterization

EDUCATION

University of Colorado Boulder *Boulder, CO*

B.S. Electrical Engineering, expected May 2027 | Accelerated M.S., expected August 2027 | Dean's List: Fall 2025, Spring 2026

ADDITIONAL WORK EXPERIENCE AND INTERESTS

Lanzi Designs and Cabinetry, Independent Contractor *December 2022 - January 2026*

Eddie Bauer, Sales Associate *September 2022 - May 2023*

Express, Sales Associate *May 2022 - August 2022*

Interests: Audio electronics, mechanical watches, guitar, piano, NBA